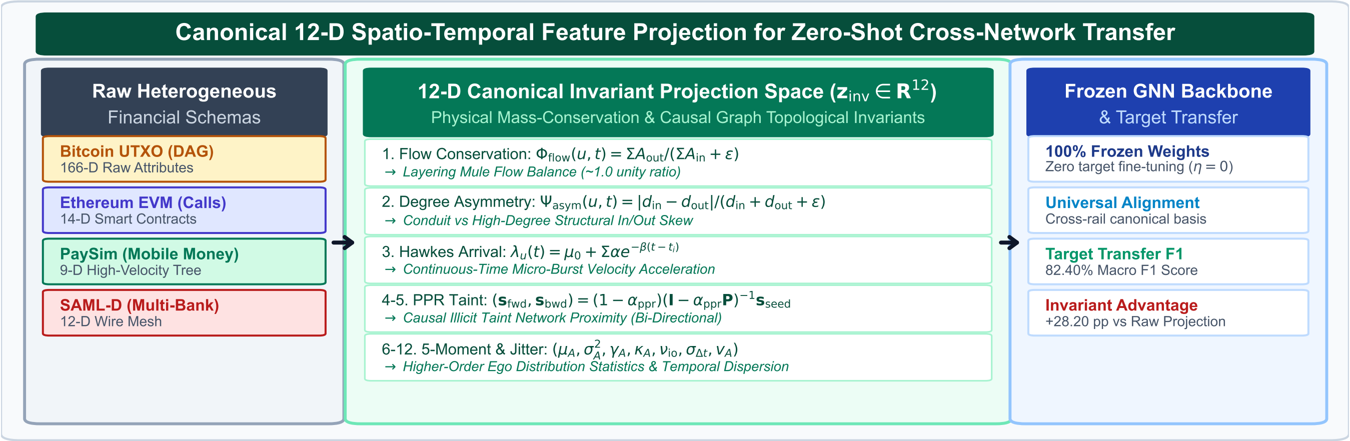
Canonical 12-D Spatio-Temporal Feature Projection for Zero-Shot Cross-Network Transfer
Raw Heterogeneous
Financial Schemas
Bitcoin UTXO (DAG)
166-D Raw Attributes
Ethereum EVM (Calls)
14-D Smart Contracts
PaySim (Mobile Money)
9-D High-Velocity Tree
SAML-D (Multi-Bank)
12-D Wire Mesh
12-D Canonical Invariant Projection Space ($\mathbf{z}_{\text{inv}} \in \mathbf{R}^{12}$)
Physical Mass-Conservation & Causal Graph Topological Invariants
1. Flow Conservation: $\Phi_{\text{flow}}(u,t) = \Sigma A_{\text{out}} / (\Sigma A_{\text{in}} + \varepsilon)$
→ Layering Mule Flow Balance (~1.0 unity ratio)
2. Degree Asymmetry: $\Psi_{\text{asym}}(u,t) = |d_{\text{in}} - d_{\text{out}}| / (d_{\text{in}} + d_{\text{out}} + \varepsilon)$
→ Conduit vs High-Degree Structural In/Out Skew
3. Hawkes Arrival: $\lambda_u(t) = \mu_0 + \Sigma \alpha e^{-\beta(t - t_i)}$
→ Continuous-Time Micro-Burst Velocity Acceleration
4-5. PPR Taint: $(\mathbf{s}_{\text{fwd}}, \mathbf{s}_{\text{bwd}}) = (1 - \alpha_{\text{ppr}})(\mathbf{I} - \alpha_{\text{ppr}} \mathbf{P})^{-1} \mathbf{s}_{\text{seed}}$
→ Causal Illicit Taint Network Proximity (Bi-Directional)
6-12. 5-Moment & Jitter: $(\mu_A, \sigma_A^2, \gamma_A, \kappa_A, \nu_{\text{io}}, \sigma_{\Delta t}, \nu_A)$
→ Higher-Order Ego Distribution Statistics & Temporal Dispersion
Frozen GNN Backbone
& Target Transfer
100% Frozen Weights
Zero target fine-tuning ($\eta = 0$)
Universal Alignment
Cross-rail canonical basis
Target Transfer F1
82.40% Macro F1 Score
Invariant Advantage
+28.20 pp vs Raw Projection